

Predicting chaperone-interactome responses from protein sequence: salt, heat and staurosporine

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Summary

Two mass-spectrometry pull-down experiments were used to ask how well a protein’s own sequence predicts how its association with a chaperone changes under a perturbation. Each protein is represented by the ESMC-6B protein language model, and its log2 fold change is predicted on proteins whose sequence families were held out of training.

Response	Pull-down	Ceiling	Best sequence model	Composition alone
Salt, 150 vs 0 mM	HSPB1	99.3 %	20.4 %	7.5 %
Salt, 75 vs 0 mM	HSPB1	98.6 %	17.2 %	6.2 %
Heat, 43 vs 35 °C	DNAJA1, DNAJB11	99.2 %	27.5 %	8.9 %
Heat, 37 vs 35 °C	DNAJA1, DNAJB11	86.5 %	12.2 %	3.5 %
Staurosporine, 37 °C	DNAJA1, DNAJB11	75.5 %	6.5 %	0.3 %

All values are percent of variance explained in held-out proteins. The ceiling is the share of the variance that is reproducible between independent sets of biological replicates, and so the most any predictor could explain.

- **Sequence predicts all three responses, and the heat response best.** The embedding explains 2.3 times or more what amino-acid composition does, and already contains composition.
- **The responses are far from saturated.** The best models reach 6-28 % of the variance against ceilings of 75-99 %, and the gap is not measurement noise.
- **A nonlinear readout helps where the signal is broad.** An ensemble of small neural networks averaged with linear regression beats linear regression alone on salt and on heat, but not on staurosporine, whose signal is concentrated in a few kinase families.
- **Staurosporine acts on protein kinases.** Kinases are specifically depleted from both co-chaperone pull-downs ($p = 2e-20$), while some, notably the PKC family, rise. The 3.6 % of proteins that are kinases carry about half of the response’s variance.
- **One nonlinear model serves several temperatures.** For heat, a network trained at 43 °C needs only a new linear readout to predict 37 °C. For staurosporine, a network trained at another temperature does as well as or better than the temperature’s own.
- **Part of what is predicted is the family.** Half to three quarters of the model’s success on proteins in large families is getting the family’s average response right.

Data and approach

Salt titration. HSPB1 pull-downs from 7,184 proteins at 0, 75 and 150 mM salt, each as 2 technical x 5 biological replicates of log2 intensities. The fold change is the difference of biological-replicate means, kept when both conditions have at least 7 of 10 values (6,989 proteins at 75 mM, 6,695 at 150 mM).

GA_33. DNAJA1 and DNAJB11 pull-downs at 35, 37 and 43 °C, with and without 10 μ M staurosporine, 6 biological replicates each (9,871 protein groups). Each run is median-normalised, so fold changes are relative to the pull-down as a whole. Heat responses are measured without staurosporine; staurosporine responses are measured at each temperature. Proteins are kept when both groups have at least 4 of 6 values. The two co-chaperones show essentially the same heat and staurosporine responses (correlations 0.83-1.0 after correcting for noise), so their average is used as the main target: the same quantity measured with twice the replicates.

Ceiling. For each fold change, the biological replicates are split into two disjoint halves, the fold change is computed from each, and the agreement is corrected to the full replicate count. This reliability is the largest fraction of variance any predictor could explain.

Prediction. Each protein’s sequence (all but two digestion enzymes resolved from UniProt) is embedded with ESMC-6B and averaged over residues; long proteins are embedded in overlapping windows. Proteins are clustered at 30 % sequence identity, and every model is scored by 5-fold cross-validation in which whole clusters are held out, so predictions are never made from a close relative. Models: ridge regression, and an ensemble of 16 two-layer neural networks selected on an inner held-out set of clusters. Intervals are 95 % bootstrap intervals over sequence clusters.

Salt sensitivity (HSPB1)

Salt weakens electrostatic interactions, so the fold change measures how salt-sensitive each protein’s association with HSPB1 is. A typical protein moves 1.6-fold at 75 mM and 1.9-fold at 150 mM, and these changes are highly reproducible: 99 % of the variance is explainable in principle.

What sequence explains. Each row adds features to the one above.

Features	75 mM	150 mM
Protein length	0.0 %	0.0 %
Amino-acid composition (20 fractions)	6.2 %	7.5 %
+ length	6.4 %	8.0 %
+ charge, hydrophobicity, charge patterning	6.6 %	8.2 %
ESMC embedding (final layer)	15.3 %	17.0 %
ESMC embedding + composition	15.3 %	17.1 %
<i>Measurement ceiling</i>	<i>98.6 %</i>	<i>99.3 %</i>

Composition alone explains 6-8 %. Charge and hydrophobicity add almost nothing beyond it, which for a salt-disruption experiment is notable: the obvious electrostatic descriptors carry no information that the raw composition lacks. The embedding adds a further +9.1 points at 75 mM [7.7, 10.5] and +9.6 at 150 mM [8.2, 10.9], and appending composition to it gains nothing, so the embedding contains composition and more. Length explains nothing. Nearest-neighbour lookup in the same embedding explains only 6-7 %, so the signal is not simply family resemblance.

Which layer. Of nine evenly spaced layers of the 81, layer 50 predicts salt sensitivity best (16.0 % and 18.2 %), about one point above the final layer; the deepest layers track the protein’s abundance rather than its salt response.

Linear versus nonlinear readout (ESMC layer 50):

	75 mM	150 mM
Ridge regression	16.0 %	18.2 %
Neural-network ensemble	16.5 %	20.3 %
Ensemble and ridge averaged	17.2 %	20.4 %
Gain of the average over ridge	+1.2 [+0.6, +1.7]	+2.2 [+1.6, +2.8]

The ensemble alone gains +0.5 [-0.7, +1.7] points at 75 mM and +2.2 [+1.0, +3.4] at 150 mM. Averaging it with ridge is positive at both doses. Fitting the two doses jointly, although they correlate at 0.91, gains nothing over fitting them separately.

Heat response (DNAJA1 and DNAJB11)

Raising the temperature from 35 to 43 °C changes what both co-chaperones pull down by about 2.2-fold for a typical protein, almost free of noise (2 % of the variance), and the two co-chaperones respond alike (correlation 0.93).

Target	Ceiling	Composition	ESMC, layer 50	ESMC, final layer	Ensemble + ridge
43 vs 35 °C, averaged	99.2 %	8.9 %	22.5 %	25.7 %	27.5 %
43 vs 35 °C, DNAJA1	97.9 %	10.1 %	23.0 %	26.0 %	27.7 %
43 vs 35 °C, DNAJB11	98.3 %	8.3 %	21.4 %	25.0 %	27.2 %
37 vs 35 °C, averaged	86.5 %	3.5 %	10.8 %	11.4 %	12.2 %

This is the most sequence-predictable response in either experiment. The embedding adds +13.6 points over composition for the averaged 43 °C response [12.3, 14.9], and here the final layer is better than layer 50, by 3-4 points: the best layer depends on the property being predicted. Averaging a neural-network ensemble with ridge beats ridge on all four heat targets, by +1.8 [+1.2, +2.4], +2.1 [+1.6, +2.6], +2.5 [+1.9, +3.0] and +0.8 [+0.5, +1.1] points. Which proteins a co-chaperone binds more of at 43 °C plausibly depends on thermal stability and aggregation propensity, both encoded in sequence and structure.

Staurosporine response

10 μ M staurosporine changes what the co-chaperones pull down by only about 1.15-fold, and half of that is replicate noise. The ceiling for a single co-chaperone is 49-64 %; averaging the two raises it to 71-76 %.

Target (both co-chaperones)	Ceiling	Composition	ESMC, layer 50
35 °C	71.7 %	1.6 %	6.0 %
37 °C	75.5 %	0.3 %	6.5 %
43 °C	71.4 %	0.5 %	5.7 %

Staurosporine acts on protein kinases. Staurosporine is an ATP-competitive inhibitor of most protein kinases, and the 346 kinases detected shift as a class. The 673 other ATP-binding proteins do not, and the heat response shows no kinase signal at all.

	Kinases	Other ATP-binding	All others	Kinase shift	p	Depleted 5 %
35 °C	-0.15	+0.01	+0.00	-0.73 sd	2e-7	6.0x
37 °C	-0.19	-0.01	+0.02	-1.05 sd	2e-20	6.5x
43 °C	-0.20	-0.01	+0.00	-1.12 sd	5e-13	5.1x
<i>Heat (control)</i>	<i>+0.11</i>	<i>+0.33</i>	<i>+0.14</i>	<i>-0.03 sd</i>	<i>0.56</i>	<i>1.3x</i>

Mean log2 fold change. Kinase shift: kinases minus all others, in standard deviations of the fold change; p from a Mann-Whitney test. Depleted 5 %: how over-represented kinases are among the 5 % most depleted proteins. Heat (control): the same comparison for the 43 vs 35 °C heat response.

The most depleted kinases include FER, CAMK1, CAMKK2, CSK, CDK5, CHEK2, CDK2 and STK4 (down 2-

to 6-fold). Kinases are also over-represented among the most enriched proteins: the PKC isoforms, PRKG1 and the PKN kinases rise. Staurosporine therefore shifts kinase-chaperone association in both directions, kinase by kinase.

Kinases dominate the response. The 295 kinases are 3.6 % of the proteins but, because their changes are so large, carry most of the variance, and most of what the model explains comes from them:

Staurosporine	Variance in kinases	Explained, from kinases	R2 within kinases	R2, all others
35 °C	44 %	50 %	3.5 %	5.8 %
37 °C	54 %	62 %	1.5 %	6.6 %
43 °C	63 %	81 %	1.3 %	3.2 %

The model captures the kinases mainly by knowing that kinases as a class go down. Which kinases go down and which go up it barely predicts (1-3.5 % within the kinases); among all other proteins it explains 3-7 % of a much smaller variance.

No nonlinear gain. Unlike salt and heat, the neural-network readout does not beat ridge reliably here (-13, -2 and +7 points at the three temperatures). Because the signal sits in a few kinase families, which families fall in the selection set decides which model looks best. The 43 °C gain comes almost entirely from the PKC family, whose held-out isoforms the network predicts from their relatives. Ridge regression remains the model to use for this response.

The same proteins at every temperature?

For each pair of responses on the same proteins: how similar the true responses are, how similar the model's predictions are, and whether the same proteins are well explained. A protein's share of the explained variance is how much the model reduces its squared deviation from the mean; these shares add up to R2.

Pair	r, responses	r, predictions	r, explained	Top-5 % overlap	R2	R2 from other
Salt, 75 / 150 mM	0.91	0.91	0.76	14x	20.4 %	17.8 %
Heat, 37 / 43 °C	0.68	0.80	0.49	6.3x	27.1 %	17.0 %
Staurosporine, 35 / 37 °C	0.54	0.40	0.12	6.9x	8.3 %	1.6 %
Staurosporine, 37 / 43 °C	0.44	0.40	0.07	6.5x	7.1 %	1.9 %
Staurosporine, 35 / 43 °C	0.47	0.72	0.18	7.4x	7.1 %	6.1 %

r, explained: correlation of the proteins' shares of explained variance. Top-5 % overlap: how often the 5 % best-explained proteins coincide, relative to chance. R2 is for the second response of each pair; "R2 from other" scores the first response's predictions, rescaled linearly, against it.

- **Salt:** the same proteins at both doses. The 75 mM predictions alone explain 17.8 % of the 150 mM response.
- **Heat:** largely the same proteins. The predictions at 37 and 43 °C correlate at 0.80, the best-explained 5 % overlap six-fold beyond chance, and the 37 °C predictions explain 17 % of the 43 °C response, against 27 % for its own model.
- **Staurosporine:** different proteins at each temperature, apart from a small shared core. The responses correlate at 0.44-0.54, but which proteins the model explains barely carries over (0.07-0.18), and one temperature's predictions explain only 1.6-6 % of another's.

One nonlinear model, different linear readouts?

Two tests, on the proteins kept at every temperature, with a fixed network configuration:

- **One network with a separate linear output per temperature**, trained jointly, against separate networks.
- **A network trained on one temperature, frozen, with a new linear readout fitted for another.** Each is compared with the second temperature’s own network, read out the same way.

Shared vs separate networks		Network from the other temperature vs own	
Heat, 37 vs 35 °C	+0.1 [-0.2, +0.5]		from 43 °C: +0.4 [-0.6, +1.4]
Heat, 43 vs 35 °C	-0.4 [-0.8, +0.0]		from 37 °C: -3.4 [-4.7, -2.1]
Staurosporine, 35 °C	-0.8 [-1.8, +0.3]	from 37 °C: -0.4 [-1.8, +1.3];	from 43 °C: +1.5 [-0.4, +3.6]
Staurosporine, 37 °C	+0.2 [-0.5, +1.0]	from 35 °C: +1.6 [+0.4, +3.1];	from 43 °C: +3.5 [+0.9, +6.4]
Staurosporine, 43 °C	+2.8 [-0.5, +6.2]	from 35 °C: -0.5 [-5.2, +4.9];	from 37 °C: +0.0 [-3.8, +4.5]

Change in percent of variance explained, with 95 % intervals over sequence clusters.

For heat, yes, one nonlinear model serves both temperatures. A single network with two linear outputs matches separate networks to within 0.4 points. A network trained only on the 43 °C response, frozen and given a new linear readout, predicts the 37 °C response as well as a network trained on 37 °C. The reverse fails: the 37 °C network loses 3.4 points on the 43 °C response. The 43 °C response contains the 37 °C pattern and more.

For staurosporine, a network from another temperature is never worse and sometimes better. For 37 °C, networks trained at 35 or 43 °C beat 37 °C’s own, by 1.6 and 3.5 points. That is the pattern expected if one sequence determinant underlies all three temperatures and each temperature measures it with independent noise. The per-protein differences between temperatures above then reflect that noise more than different biology.

Protein families

Proteins were grouped by UniProt family (top level; families of at least 15 detected members), and by subcellular location.

	Salt, 150 mM	Heat, 43 vs 35 °C	Staurosporine, 37 °C
Proteins in families of 15 or more	972 (23 families)	1,327 (34)	1,308 (34)
Share of the true response that lies between families	16 %	20 %	6 %
Model, R2 on these proteins	20.6 %	29.3 %	6.5 %
R2 from each family’s average prediction alone	13.6 %	14.7 %	4.9 %
Correlation within families	0.29	0.43	0.13

For proteins in large families, half to three quarters of what the model explains comes from getting each family’s average response right; the rest comes from ranking proteins within a family, which works best for heat ($r = 0.43$).

Best and worst predicted families (share of their deviation from the overall mean explained):

Response	Best predicted	Worst predicted
Salt, 150 mM	mitochondrial carriers (88 %, up), KRAB zinc fingers (49 %, up), class I aminoacyl-tRNA synthetases (52 %, down), ABC transporters (42 %)	small GTPases, DEAD-box and other helicases, myosins and kinesins (below 0 %)

Response	Best predicted	Worst predicted
Heat, 43 vs 35 °C	mitochondrial carriers (77 %, down), MFS and P-type transporters (67 %, down), deubiquitinases (57 %, up; within-family $r = 0.79$), dynamins (53 %, up), class II aminoacyl-tRNA synthetases (48 %, up)	tubulins (true down, predicted up), cyclins, actins, ABC transporters
Staurosporine, 37 °C	protein kinases (3.6 % of proteins, 62 % of what is explained), ATP-dependent AMP-binding enzymes, translation GTPases, AAA ATPases	dynamins, SAM methyltransferases, actins

Membrane transporters stand out in both salt and heat, in opposite directions: they rise in the HSPB1 pull-down with salt and fall in the co-chaperone pull-downs with heat, and in both the model predicts the family average well. Cytoskeletal structural proteins (tubulin, actin) are the most consistently mispredicted.

By compartment the heat response is predicted about equally well everywhere (20-30 %, lowest for the cytoskeleton). For staurosporine, endoplasmic-reticulum proteins are predicted best (17 %, within $r = 0.41$) and secreted proteins worst (about 0 %). For salt, mitochondrial, Golgi and membrane proteins (25-28 %) are predicted better than cytoplasmic and cytoskeletal ones (17 %).

Using individual samples instead of replicate means

Each protein’s measurements were also modelled sample by sample, with a separate noise level for each run and a noise level that rises for low-abundance proteins (4-fold from the most to the least abundant fifth). Three ways of letting that noise discount samples were compared against the plain replicate mean, in the same networks:

	Salt	Heat	Staurosporine
Measurement noise, share of variance	0.4-0.8 %	2 %	15 %
Weight by measurement noise alone	-4.1 to -1.3	-1.9 to -0.2	-3.7 to +7.7
Full likelihood (model error + noise)	-0.3 to +0.2	-0.2 to +0.3	-2.1 to +10.8
Network predicts its own uncertainty	-0.2 to +0.6	-0.2 to +0.5	-1.8 to +15.1

Change in percent of variance explained, against the same network trained on the replicate mean.

Weighting by measurement noise alone hurts, because it gives the best-measured proteins up to 16 times the influence and those are not the easiest to predict. Under the full likelihood the model’s own error (27-48 % of the variance on the salt data) dwarfs measurement noise, so the weights come out nearly uniform and the result equals the plain mean. Staurosporine, where noise is largest, shows a slightly positive median effect, but its spread is far wider than the effect.

Conclusions and next steps

Protein sequence, read through a language-model embedding, predicts how chaperone association responds to salt, heat and staurosporine, in each case well beyond amino-acid composition and in proteins with no close relative in the training data. Heat is the most predictable (27.5 % of a 99 % ceiling); staurosporine is the least, and its signal is dominated by protein kinases.

1. **Input lysate.** Measuring each pull-down against its own input would separate a change in a protein's amount from a change in its binding, the open question for the depleted kinases.
2. **Staurosporine affinity across the kinome.** Measured binding constants are the natural test of which kinases go down and which go up.
3. **Several layers at once.** The best embedding layer differs by target by 3-4 points.